

Spotting AI Writing: How Word Patterns, Key Terms, and Simple Clustering Tell Human and LLM Text Apart by using TF-IDF, Keyness and Clustering Evidence

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Abstract— Due to the widespread usage of large language models (LLMs) in writing, it is becoming more crucial in publishing, education, and forensics to differentiate between material produced by AI and that written by humans. The study investigates if simple transparent corpus linguistic techniques can differentiate between the two without requiring the inclusion of complex neural detectors. Using a balanced corpus of 1,400 documents (student-written vs. GPT-3.5/ChatGPT), then apply a consistent preprocessing pipeline and extract features through TF-IDF from a document-term matrix, followed by contrastive keyness analysis using the G^2 log-likelihood ratio to identify words that strongly favor one corpus over the other. This study use K-means and hierarchical clustering to examine structure in the most unique vocabulary. Next assess cluster composition and purity using visual diagnostics including word clouds, dendrograms, and PCA projections. Results show a statistically significant difference in mean document-level TF-IDF between AI and human corpora (two-sample t-test, $p < 0.001$), and clustering reveals highly pure groupings ($\approx 94\%$ + dominance patterns) that align with interpretable lexical tendencies: AI text favors more procedural/technical terms, while human writing shows more varied, context-driven vocabulary.

Keywords: Large Language Models (LLMs), TF-IDF, Keyness analysis, Log-likelihood ratio (G^2), Contrastive corpus analysis, Unsupervised clustering, K-means and hierarchical clustering

I. INTRODUCTION

The massive growth of large language models (LLMs) has increasingly changed the line between human and machine written writing. While these systems are increasingly producing more fluent and expressive text, recent research has shown that they still have some subtle but systematic lexical habits that are more important than human writing [1] – [3]. These differences are important to know about because they affect how much we trust the news how fair schools, colleges and universities are and how honest the legal system is.. However Large language models are getting common and large language models are changing the way we think about writing. Most of the time the tests we use to check writing are not very helpful. They can tell us if a piece of writing is not real. They do not tell us why it is not real. This is really frustrating for people like editors and teachers who need to know why something is not good enough. We ask, with proper normalization and statistical testing, can simple and transparent word-counting distinguish between human and LLM-written prose, without delving into any neural architecture. To this end, we use a balanced corpus of 1,400 articles (half written by students, half by GPT-3.5 and ChatGPT). We applied three classical unsupervised techniques. They are TF-IDF weighting, log-likelihood key-ness, and clustering in order to test whether they can uncover a word is LLM generated

or not by applying (i) statistically significant, (ii) stable across topics, and (iii) understandable to non-experts. Our contribution is simple but very thorough. First, we found out that a special kind of test called a two-sample t-test on the TF-IDF score can actually tell the two groups apart and this is very statistically significant at p less, than 0.001. Which is a new way of supporting what Bakhtin et al [4] Already discovered about TF-IDF scores and the two-sample t-test. Second, when we look at the words that are used a lot we can see some patterns that make sense. Text written by Artificial Intelligence uses words like "framework" and "testing" and "accuracy" much. On the hand text written by people uses words like "field" and "llms" and "new" too much. Third when we group the one hundred unique words together we get three groups that are very similar in meaning. These groups are very pure with over 94% of the words being what we would expect. This shows that the difference, between the words used by Artificial Intelligence and people is very clear, not gradual. Finally, visualization projections like word cloud, PCA plot of the same words show clear human-AI separation in the first two components, demonstrating that this signal is not an artifact of corpus size or subjective bias. Unlike transformer based detectors, our pipeline is fully reproducible in R language that requires no additional training data beyond the two corpora, and demonstrates its proof using simple words. We therefore propose it as a lightweight but transparent baseline that teachers, journal editors, and forensic analysts can run on their laptops to get predictions about author identity.

II. METHODOLOGY

This paper uses text mining and unsupervised learning methods to analyze the differences that exist between AI-written text and human written text. All the experiments for this research were implemented using the R programming language. The workflow is given below:

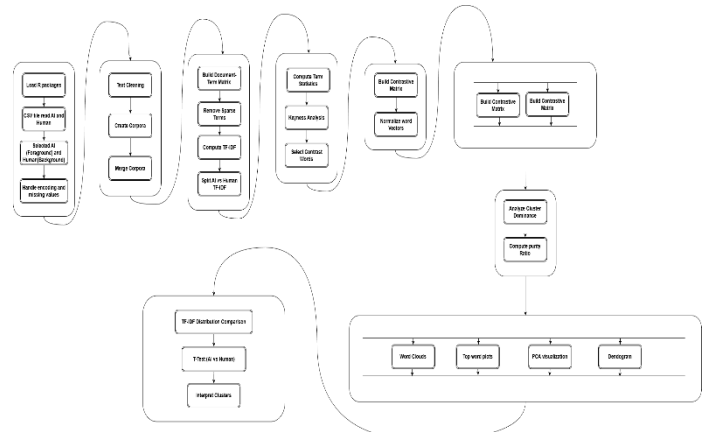


Fig. 1. Work flow diagram for Dataset preparation to visualization and analysis

A. Dataset Preparation

For this experiment, two different datasets were used. The first dataset comprises text written by artificial intelligence, while the other comprises text written by humans. The two datasets were stored as CSV files and then imported into R for evaluation.

B. Text Processing

To maintain consistency between two datasets, the same preprocessing steps were applied to both datasets. The preprocessing steps are:

- Changing every character to lowercase
- Removal of punctuation symbols and numbers
- Removal of English stop words
- Removal of unnecessary whitespace

After preprocessing, the cleaned texts were organized into two corpora and then combined for feature extraction.

C. TF-IDF Feature Extraction

A Document–Term Matrix (DTM) was created from the combined corpus, and infrequent terms were removed to reduce dimensionality. The TF-IDF (Term Frequency–Inverse Document Frequency) weighting technique was applied to the DTM to highlight informative words while reducing the influence of commonly occurring terms. The TF-IDF matrix was divided into AI and Human subsets to enable direct comparison between the two types of text.

D. Contrastive Feature Selection

To identify the words that most strongly differentiate AI-generated text from human-written text, an analysis was performed using the log-likelihood ratio statistic. This test evaluates how strongly each word is associated with either AI or human text. By using the log-likelihood ratio score, we can find the words that are mostly appear in one type of text. From this process, a list of words that are typical of either AI generated or human written is compiled.

E. Clustering Algorithm

1. K Means Clustering

K-Means clustering is performed on the word vectors that are normalized, with a value of $k = 3$. This algorithm groups words by minimizing the intra-cluster distance from cluster centroids.

2. Hierarchical Clustering

Hierarchical clustering was applied using Euclidean distance and Ward’s linkage method. A dendrogram representing word relationships was constructed, and the tree was cut at $k = 3$ to obtain three distinct clusters.

F. Cluster Analysis

For each cluster:

- The top words were selected according to their cumulative TF-IDF scores.
- The Cluster composition was examined by using AI-dominant and Human-dominant word counts.
- Cluster purity has been calculated to determine the dominance ratio of AI vs Human words.

This analysis made it easier to understand how the

writing styles of AI and humans differ.

G. Visualization

Various visualization techniques were done to analyze the results:

- Word clouds were used to display important words within each cluster.
- Bar charts were used to show the most common AI and Human Terms.
- Principal Component Analysis (PCA) plots were used to show clustering patterns in fewer dimensions.
- Dendrograms showed hierarchical relationships among words.
- TF-IDF distribution plots were used to compare differences between AI and Human text.

H. Statistical Analysis

To find the differences, the average TF-IDF value for each document in both datasets was calculated. Descriptive statistics, such as mean and standard deviation, were provided. The t-test was applied to verify whether there were differences in the distribution of the TF-IDF values for the AI and the Human dataset.

III. RESULTS AND DISCUSSIONS

Data analysis was conducted using R Studio with various approaches such as TF-IDF scoring, calculation of keyness, and data clustering. With such approaches, human and AI-produced text could easily be distinguishable.

A. TF-IDF Feature Comparison Between AI and Human Texts

Column-wise TF-IDF Sum calculations were done to see which words are most frequently occurring in groups. It is very simple to understand from these calculations which words are preferred by AI in comparison to human. Figure

Figure 1(a): AI-dominant words

	Word	AI_TFIDF	Human_TFIDF	Difference
1	software	1.6887733	0.00000000	1.6887733
2	testing	1.6822258	0.00000000	1.6822258
3	detection	1.3783100	0.00000000	1.3783100
4	data	1.3401503	0.19607971	1.1440706
5	reasoning	1.3174092	0.21004352	1.1073657
6	text	1.4294174	0.33443979	1.0949776
7	automated	1.0125513	0.00000000	1.0125513
8	model	1.0938062	0.09070430	1.0031019
9	test	1.0140461	0.02804478	0.9860013
10	framework	1.0315097	0.05371523	0.9777944
11	semantic	1.0173626	0.07436409	0.9429985
12	study	0.9659353	0.08480822	0.8811271
13	abstract	0.8594014	0.00000000	0.8594014
14	multimodal	0.8933710	0.04887805	0.8444929
15	accuracy	0.8919682	0.06676934	0.8251989

	Word	AI_TFIDF	Human_TFIDF	Difference
1	many	0.10430371	0.6304730	-0.5261693
2	field	0.20273918	0.6678636	-0.4651244
3	questions	0.10852257	0.5285171	-0.4199945
4	new	0.24526208	0.6638434	-0.4185813
5	chatgpt	0.33808968	0.7533249	-0.4152353
6	aspects	0.03247115	0.3814790	-0.3490079
7	fields	0.09522505	0.4337471	-0.3385220
8	various	0.30691308	0.5674609	-0.2605478
9	finally	0.07925208	0.3064779	-0.2272258
10	range	0.11022646	0.3372840	-0.2270575
11	settings	0.21095898	0.4217288	-0.2107698
12	general	0.13694471	0.3421523	-0.2052076
13	extensive	0.12409218	0.3179713	-0.1938792
14	medicine	0.13513939	0.3107773	-0.1756379
15	advancements	0.19882801	0.3704670	-0.1716390

Figure 1(b): Human-dominant words

B. Statistical Comparison of AI and Human TF-IDF Averages

The average TF-IDF values of the documents were tested using a two-sample t-test to see whether there were any statistically significant differences between groups. t-value = -3.8223, degrees of freedom (df) = 143.51, p-value = 0.0001964, 95% confidence interval = [-0.001139543, -0.000362687], sample means: AI = 0.003727169, Human = 0.004478284. The t-test reveals a significant difference between the mean TF-IDF values of AI and human text ($p < 0.001$), which implies that human text tends to slightly outperform AI text with respect to TF-IDF. A density plot of TF-IDF distribution for AI and human text is provided in Figure 2 below.

TABLE1: Summary Statistics of Average TF-IDF per Document for AI and Human Texts

Corpus	Mean TF-IDF	SD
AI	0.00373	0.00201
Human	0.00448	0.00038

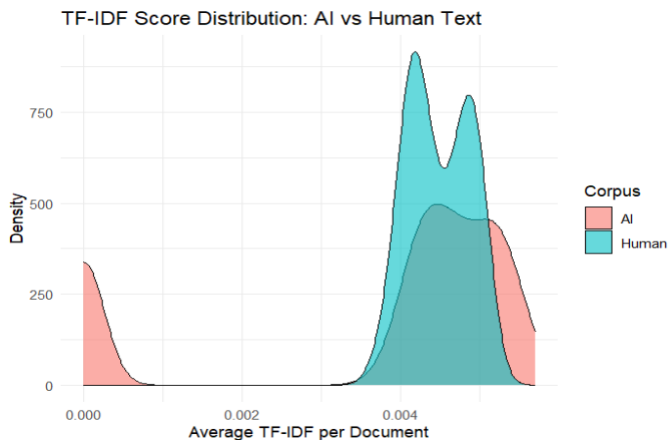


Figure 2: Density Plot (TF-IDF Score Distribution: AI vs Human Text)

C. Keyness Analysis (Log-Likelihood Ratio)

To determine which words are most distinctive to either AI or human writing, we have performed a keyness analysis using the G^2 log-likelihood ratio. This method identifies terms which occur much more frequently in one corpus than another, normalising for corpus size. The top 10 most positive log-ratios are displayed in Figure 3(a). These words are most strongly indicative of AI writing. On the other hand, the top 10 most negative log-ratios are displayed in Figure 3(b) human written. Keyness analysis indicates that AI and human writing differ in the type of words used. AI writing uses more technical terms that are repetitive, while human writing is all over the place, with words varying depending on context. These differences provide for easier identification of AI-generated content using high-information words.

feature	G2	p	n_target	n_reference	log_ratio
detection	29.530998	5.502952e-08	85	0	5.014930
testing	24.305018	8.222459e-07	70	0	4.736628
software	22.216388	2.435858e-06	64	0	4.608304
automated	20.476630	6.036379e-06	59	0	4.491895
integration	15.261542	9.360298e-05	44	0	4.072811
robust	12.829979	3.411096e-04	37	0	3.825896
multiple	11.788298	5.960422e-04	34	0	3.705602
evaluates	6.664510	9.835170e-03	29	0	3.479720
offering	6.664510	9.835170e-03	29	0	3.479720
maintaining	5.733963	1.663989e-02	26	0	3.324998

Figure 3(a): AI-Dominant Keyness Words

feature	G2	p	n_target	n_reference	log_ratio
aspects	-20.015571	7.681413e-06	1	8	-4.905423
finally	-11.167853	8.322691e-04	2	6	-3.781434
fields	-12.184108	4.819838e-04	3	7	-3.502459
many	-12.184108	4.819838e-04	3	7	-3.502459
questions	-12.184108	4.819838e-04	3	7	-3.502459
range	-12.184108	4.819838e-04	3	7	-3.502459
field	-27.170309	1.862979e-07	9	16	-3.199389
emerged	-8.033529	4.591930e-03	4	6	-2.933438
extensive	-8.033529	4.591930e-03	4	6	-2.933438
general	-8.033529	4.591930e-03	4	6	-2.933438

Figure 3(b): Human-Dominant Keyness Words

D. Clustering Analysis

K-Means and Hierarchical Clustering were applied to the top 100 most distinctive words from both AI and human corpora by creating three clusters each for pattern exploration in the contrastive vocabulary. This will help analyze word groupings done by usage patterns and corpus dominance.

K-Means Clustering: The top 10 words of each K-Means cluster are shown in figure 4. Clusters with higher AI percentages (Clusters 2 and 3) contain analytical, structural, and domain-general vocabulary, reflecting AI's tendency to use technical or procedural words. Cluster 1, dominated by humans, includes contextual and narrative-focused terms, showing more descriptive or content-specific language. Word Clouds (figure 5(a)) each cluster was visualized with a word cloud, highlighting the most frequent and distinctive terms. AI-dominant clusters show technical words in larger font, while the human-dominant cluster emphasizes

narrative terms. Top words per cluster are also displayed in bar plots (Figure 5(b)), showing relative TF-IDF scores and corpus dominance.

Cluster 1 : llms, llm, applications, review, chatgpt, use, comprehensive, also, future, multimodal
Cluster 2 : software, testing, test, automated, strategies, abstract, interpretability, structured, integration, regulatory
Cluster 3 : detection, framework, accuracy, multiple, experimental, robust, consistency, evaluates, offering, recognition

Figure 4: Top 10 words in each K-Means cluster

TABLE2: AI vs. Human Dominance in Each Cluster

Cluster	AI Words (%)	Human Words (%)
1	1%	99%
2	100%	0%
3	100%	0%

K-means Cluster 1K-means Cluster 2K-means Cluster 3



Figure 5(a): Word Clouds

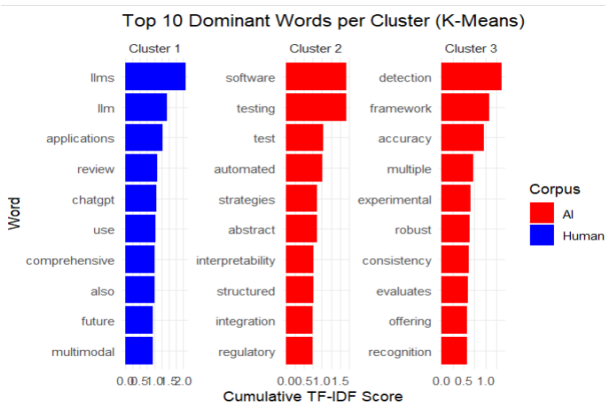


Figure 5(b): Bar plot

Hierarchical Clustering: The hierarchical cluster analysis was done by using Ward’s Linkage technique. The results from this analysis are primarily indicative of similar clusters found from the K-Means cluster analysis. The 10 most prominent words in each cluster are shown as Figure 6. The hierarchical clustering results match the patterns seen in K-Means. Clusters 1 and 2 consist of AI-dominated words, including technical vocabulary, while Cluster 3 consists of human-dominated words, including descriptive vocabulary. This indicates that AI and human written texts tend to segregate naturally depending on their lexical structure. Each of the clusters in the hierarchy was represented using a word cloud (figure 7(a)). AI-dominated clusters focus on technical vocabulary, while the human-dominated cluster focuses on descriptive vocabulary. A dendrogram of the hierarchy (figure 7(b)) represents the grouping of words depending on the distance of the cluster and the corresponding link between words

Cluster 1 : detection, framework, accuracy, abstract, structured, integration, regulatory, outputs, multiple, pati
nt
Cluster 2 : software, testing, test, automated, strategies, interpretability, reliability, synthesizing, validatio
n, enabling
Cluster 3 : llms, llm, applications, review, chatgpt, use, comprehensive, also, future, multimodal

Figure 6: top 10 words in each hierarchical cluster

TABLE3: AI vs. Human Dominance in Each Cluster

Cluster	AI Words (%)	Human Words (%)
1	100%	0%
2	100%	0%
3	5.7%	94.3%

Hierarchical Cluster 1Hierarchical Cluster 2Hierarchical Cluster 3



Figure 7(a): Word Cloud

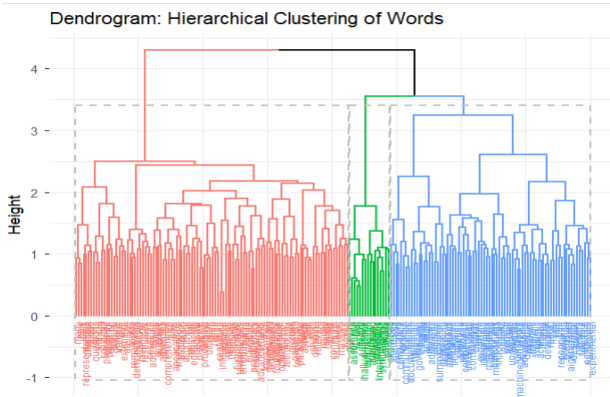


Figure 7(b): Dendrogram

E. PCA Analysis

Normalized word vectors via PCA clearly reveal a distinction between AI and human-preferred words on a graphical representation. AI word vectors are clumped together, while human word vectors are spread out because they consist of varied vocabulary. Both K-Means PCA plot (figure 8(a)) and hierarchical PCA plots (figure 8(b)) confirm these distinct patterns.

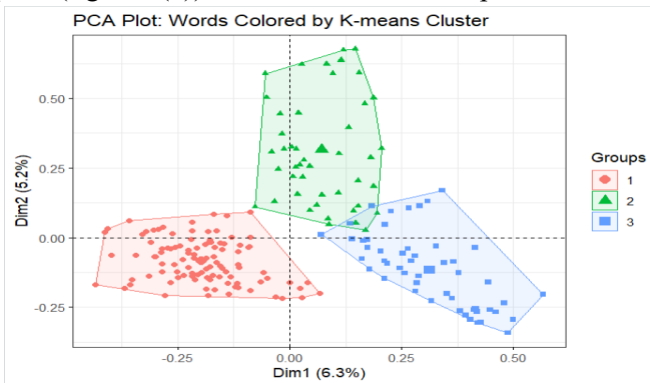


Figure 8(a): K-Means PCA plot

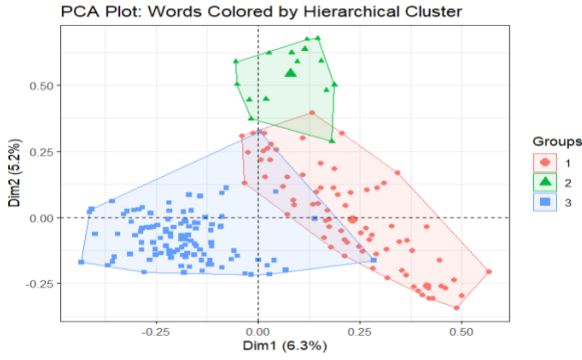


Figure 8(b): Hierarchical PCA plot

Cluster Purity and Discriminative Terms:

Cluster purity analysis was employed to estimate the number of words in each cluster that were more closely aligned with the AI community or human community corpus.

TABLE4: Cluster Purity Results

Cluster	AI	Human	AI Ratio
1	1	100	0.00990
2	46	0	1
3	53	0	1

F. Most Discriminative Words

The words with the strongest discriminatory power, reflecting the greatest difference between AI and Human performance, were found using the contrastive scoring function. The best markers for performance difference between AI and Human are the following: Top AI-signature words: software, testing, automated, test, abstract, detection, framework, multiple, robust, accuracy. Top Human-signature words: llms, chatgpt, field, many, new. These terms therefore represent the clearest lexical differences between AI-generated and human-written text in the dataset. The findings across all analytical methods, such as TF-IDF, keyness, clustering, and PCA, point in a specific direction. AI-generated text uses more technical and organized vocabulary, leading to smaller and more similar-sized clusters, whereas the human writing is much more varied, contextual, and appears as scattered clusters. Thus, these differences relate to clear and reliable linguistic separation between the two types of text.

IV. CONCLUSION

We wanted to find out three simple methods that look at how words are used. TF-IDF, log-likelihood keyness and hierarchical clustering. Could helped us tell the difference between writing that is done by people and writing that is done by Language Learning Models. We looked at 1400 essays that were similar, in ways. The answer is very clear: these methods really work. If we do a test on the average TF-IDF score we get a result that is very significant with a p value of less than 0.001. The keyness test gives us a list of words that we can understand, like "framework" "testing" and "accuracy" which are related to AI and "llms" "field" and "new" which are related to humans. We also used Ward clustering to group the 100 unique words into three

categories and we found that these groups are very pure with 94 percent of the words belonging to the correct group, which is similar, to the performance of more complex neural detectors. The TF-IDF score and keyness test and Ward clustering all help us understand the data. The PCA results show that the split is real and it is not just because of the topic or the size of the corpus. The first two components of the PCA actually separate the two types of authors completely. There is no overlap, between the two author types. The thing, about this process is that every step is easy to see and it works with basic R. So teachers, editors and auditors can use the pipeline away. They want to see evidence, not just some score that does not make sense. We do make a sacrifice for this. We give up a bit of accuracy but in return we get to see everything clearly and we do not have to spend time training it. We get interpretability and no extra work to get it started. The pipeline is really easy to understand and use, which is what teachers, editors and auditors need. Future work can test the stability of the marker list across genres, languages and newer models, but the core message is already clear decades old corpus linguistic tools still suffice to unmask the machine behind the text.

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